Project Charter v2.0

Mission Statement

Build a standalone AI-assisted alcohol label verification prototype that helps TTB compliance agents reduce repetitive manual review work by extracting text from alcohol labels, matching labels to application records, validating required fields, and generating explainable recommendations for human review.

The application assists compliance personnel but does not replace human decision-making.

Final compliance determinations remain the responsibility of the compliance agent.

Core Principles

Principle 1: Human Review Remains Authoritative

The system may recommend:

- Pass
- Fail
- Needs Review

The system never makes final regulatory decisions.

All compliance determinations remain the responsibility of the reviewing compliance agent.

Principle 2: Local-First Architecture

Core functionality must operate without reliance on external AI APIs.

Reasons:

- Government network restrictions
- Reduced latency

- Reduced operational risk
- Easier future deployment in restricted environments
- Improved reliability

The prototype should remain functional even in environments where outbound connections are heavily restricted.

Principle 3: Explainability Over Complexity

Every validation result should clearly explain:

- What was detected
- What was expected
- How the result was determined
- Why the result was generated

Users should never be forced to trust a "black box" decision.

Principle 4: Speed Matters

Target:

- Individual label processing under 5 seconds
- Progressive display of results as labels complete processing

The application should prioritize responsiveness and perceived performance.

Principle 5: Simple User Experience

A first-time user should understand how to operate the application with minimal instruction.

The interface should be:

- Simple
- Obvious
- Low-clutter
- Accessible
- Friendly to nontechnical users

Principle 6: Batch-First Workflow

The primary workflow should support processing multiple labels against application records.

Single-label verification remains available for testing and demonstration purposes but is not the primary workflow.

The prototype should support:

- 5–20 labels per batch

Future production versions may support:

- 200–500+ labels per batch

Principle 7: Conservative Decision Making

When uncertainty exists, the system should favor human review over automated decisions.

Examples:

- Low-confidence OCR results
- Ambiguous record matches
- Missing required fields
- Poor image quality

These scenarios should generate:

Needs Review

rather than incorrect pass/fail decisions.

Success Criteria

A compliance agent can:

1. Upload application records through CSV.
2. Upload one or more alcohol label images.

3. Receive verification results in under five seconds per label.
4. Review suggested record matches.
5. Understand exactly why each field passed, failed, or requires review.
6. Export results for additional review if desired.

Supported Validation Scope

Core Required Fields

The prototype validates:

- Brand Name
- Class / Type
- Alcohol Content
- Net Contents
- Government Warning

These fields represent the primary validation scope for the prototype.

Supported Optional Fields

The prototype supports:

- Bottler / Producer
- Country of Origin
- Beverage Type

These fields are included for future extensibility and may be displayed or validated when data is available.

Future Validation Scope

Future versions may include:

Beverage-Specific Validation

- Beer-specific rules
- Wine-specific rules
- Distilled spirits rules

Label Layout Validation

- Font size validation
- Font weight validation
- Government warning placement validation
- Label layout analysis

These capabilities are intentionally deferred to maintain focus on the core verification workflow.

Record Matching Policy

The prototype includes a record matching engine that attempts to associate uploaded labels with application records.

Confidence thresholds:

Auto Match

95% and above

Needs Review

70%–94%

No Match

Below 70%

Record matching recommendations are advisory only.

Final record assignment remains subject to human review.

Priority Levels

Tier 1 – Required

- CSV application record upload
- Image upload
- OCR extraction
- Text normalization
- Record matching engine
- Validation engine
- Results dashboard
- Human review workflow
- Explainable results
- Streamlit application
- Public deployment
- Documentation
- Sample labels
- Sample application records

Tier 2 – Strongly Recommended

- CSV export
- OCR text viewer
- Processing time display
- Batch progress indicators
- Sample batch datasets

Tier 3 – Time Permitting

- Additional image preprocessing
- Enhanced confidence scoring
- Manual record assignment tools
- Expanded validation coverage

Technical Direction

User Interface

Streamlit

OCR

Tesseract OCR

Text Matching

RapidFuzz

Programming Language

Python

Deployment

Render

The architecture should prioritize simplicity, maintainability, and rapid delivery over enterprise-scale complexity.

Explicit Non-Goals

The prototype will not include:

- COLAs Online integration
- Authentication system
- User management
- Database persistence
- Workflow engine
- Case management
- Enterprise architecture
- External AI API dependency
- Final compliance decision automation
- Production-scale batch processing
- Beverage-specific regulatory engines
- Font validation
- Label layout validation

These items are intentionally excluded to maintain focus on the core stakeholder requirements and prototype timeline.

Definition of Success

The project is successful if it demonstrates that local OCR, record matching, and rule-based validation can reduce repetitive compliance review work while preserving transparency, explainability, and human decision-making authority.